

Sachan Rai

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Experience

Software Engineering Intern

Jan 2026 - Apr 2026

Narrative Projects

Toronto, ON

- Designed an **inbound email parsing webhook**, routing mail to the service via an **MX record** and processing **100+** daily messages, recovering threads where the comms inbox was added later via **header and subject matching**
- Replaced an in-app **text-style messaging system** with an email-driven Comms Hub in **Node.js + TypeScript** on **Cloud Run**, AI-tagging emails to products at **90% accuracy** across **1000+ brand-owner products**
- Self-hosted the **OpenClaw AI framework** on a **GCP Compute Engine VM** to power email-driven order intake and daily inbox digests, scoped under a **dedicated service account** with **IAM-controlled SSH**

System Operations Intern

May 2025 - Aug 2025

Canadian Analytical Laboratories

Mississauga, ON

- Audited **200+ IT assets** across **4 departments** by building a standardized **ITAM tracker in Excel**, surfacing end-of-life devices and warranty coverage gaps, informing the infrastructure refresh and budget planning cycle
- Configured workstations for **lab, regulatory, QA, and admin teams** by deploying standardized Windows builds and installing **chromatography systems (Waters Empower, Agilent OpenLab)** under **GMP/Health Canada standards**
- Coordinated rollouts with scientists, IT leads, and regulatory staff by scheduling cutovers around active assay runs, delivering hardware and software upgrades with **zero downtime** across ongoing **clinical trial documentation**

Controls Engineer

Sep 2024 - Apr 2025

Waterloo Rocketry

Waterloo, ON

- Engineered a **mechanical airbrake system** in **SOLIDWORKS**, designing symmetric deployable drag panels that fit within the rocket's airframe and mass budget for a projected **20% improvement in aerodynamic efficiency**
- Designed and tested a **custom PCB airbrake control board** in **KiCAD**, integrating sensor inputs and **filtering noisy telemetry signals** in real time to deliver deployment commands during high-speed boost and coast phases
- Partnered with the **avionics and recovery subteam** to integrate the airbrake PCB with the main flight computer's data and power buses, resolving **cross-subsystem integration issues** to lock in a flight-ready revision

Projects

🔧 Embedded Irrigation Controller | C, STM32, PWM, UART, Proteus

- Developed **embedded firmware** on an **STM32 F401RE** running a 24-hour autonomous 3-zone irrigation cycle, with a **UART setup terminal**, **PWM control of DC and servo motors**, and interrupt-driven RPM measurement
- Designed the prototype schematic and PCB layout in **Proteus**, selecting components from manufacturer datasheets and implementing **3.3V/5V voltage translation** across all sensor and actuator interfaces

🔧 Dorm Defense Alarm System | C/C++, STM32, I2C, KiCad, STM32 CubeIDE

- Built a three-unit alarm system on **STM32 microcontrollers**, integrating **PIR sensors**, **MPU-6050 accelerometers**, and **HC-05 Bluetooth modules** to achieve **~90% motion detection accuracy** across a **20-meter range**
- Programmed real-time inter-board communication in **C/C++** via **STM32 CubeIDE** with **sub-100ms response times**, designing analog signal conditioning circuits to reduce sensor noise by **~60%** and stabilize **3.3V signals**

🔧 Automotive CAN Bus Simulator | C/C++, Python, ESP32, CAN, Arduino IDE

- Programmed embedded firmware on an **ESP32** with **MCP2515 controller and TJA1050 transceiver**, simulating **multiple virtual ECUs** transmitting **CAN frames** for vehicle speed, engine RPM, brake status, and fault conditions
- Built the **CAN network harness** from twisted-pair cabling, DB9 connectors, and 120Ω termination resistors, complemented by a **python-can analysis tool** for **real-time monitoring, filtering, and decoding** via USB

Education

University Of Waterloo

Waterloo, ON

Bachelor of Applied Sciences in Computer Engineering

Sep 2024 - Apr 2029

Technical Skills

Languages: C, C++, Python, Bash, MATLAB, VHDL, Verilog, Assembly

Hardware & Embedded: STM32, ESP32, Arduino, FreeRTOS, KiCad, SolidWorks, Fusion360, Quartus Prime

Protocols & Test: CAN, I2C, SPI, UART, PWM, Oscilloscope, Logic Analyzer, Multimeter, Soldering/Prototyping

Developer Tools: Linux, Git, Jira, Agile, STM32 CubeIDE, Proteus, PlatformIO, Pthreads, VS Code